

Manual Change History (4021528) K38, K50, QSK38, and QSK50 Service Manual

Date	24-Aug-2010
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	28-001-024-tr (/qs3/pubsys2/xml/en/procedures/28/28-001-024-tr.html)
Procedure Title	Crankshaft Seal, Rear
Update Reason	Procedure updated to include new rear crankshaft seal installer used in remove and install steps.
Date	25-Aug-2010
Section	Section 5 - Fuel System - Group 05
Procedure Number	101-005-232 (/qs3/pubsys2/xml/en/procedures/101/101-005-232.html)
Procedure Title	Fuel Pump Pressurizing Assembly
Update Reason	
Date	13-Sep-2010
Section	Section 8 - Cooling System - Group 08
Procedure Number	28-008-013-tr (/qs3/pubsys2/xml/en/procedures/28/28-008-013-tr.html)
Procedure Title	Coolant Thermostat
Update Reason	Added Notes for thermostat insert on Keel-Cooled Marine QSK50 CM2150 engines.
Date	16-Sep-2010
Section	Section F - Familiarization
Procedure Number	28-200-003 (/qs3/pubsys2/xml/en/procedures/28/28-200-003.html)
Procedure Title	Flow Diagram - Cooling System
Update Reason	Flow diagram captions were updated and a Note added about keel cooled marine thermostat flow.

Date	23-Jun-2010
Section	Section 6 - Injectors and Fuel Lines - Group 06
Procedure Number	28-006-026-tr (/qs3/pubsys2/xml/en/procedures/28/28-006-026-tr.html)
Procedure Title	Injector
Update Reason	Updated Initial Check section with new service tool.

Date	19-Feb-2010
Section	Section 2 - Cylinder Head - Group 02
Procedure Number	28-002-004-tr (/qs3/pubsys2/xml/en/procedures/28/28-002-004-tr.html)
Procedure Title	Cylinder Head
Update Reason	Updated technical content of Vacuum Test section.

Date	23-Jun-2010
Section	Section 6 - Injectors and Fuel Lines - Group 06
Procedure Number	28-022-001_06 (/qs3/pubsys2/xml/en/procedures/28/28-022-001_06.html)
Procedure Title	Service Tools
Update Reason	Updated service tools with injector tester.

Date	22-Nov-2010
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	28-001-031-tr (/qs3/pubsys2/xml/en/procedures/28/28-001-031-tr.html)
Procedure Title	Gear Cover, Front
Update Reason	Added note for clarification on water pump.

Date	22-Nov-2010
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	28-001-66-tr (/qs3/pubsys2/xml/en/procedures/28/28-001-66-tr.html)
Procedure Title	Gear Cover Accessory Drive Bushing
Update Reason	Added note to procedure for clarification on water pump.

Date	08-Nov-2010
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	28-001-008 (/qs3/pubsys2/xml/en/procedures/28/28-001-008.html)
Procedure Title	Camshaft
Update Reason	Updated procedure to indicate that the run-out specification applies to all journals individually.

Date	17-Dec-2010
Section	Section F - Familiarization
Procedure Number	28-200-003 (/qs3/pubsys2/xml/en/procedures/28/28-200-003.html)
Procedure Title	Flow Diagram - Cooling System
Update Reason	Added marine gear cooler flow diagram.

Date	17-Dec-2010
Section	Section 8 - Cooling System - Group 08
Procedure Number	28-008-041-tr (/qs3/pubsys2/xml/en/procedures/28/28-008-041-tr.html)
Procedure Title	Marine gear oil cooler
Update Reason	Updated marine gear cooler graphics.

Date	17-Dec-2010
Section	Section 8 - Cooling System - Group 08
Procedure Number	28-008-137-tr (/qs3/pubsys2/xml/en/procedures/28/28-008-137-tr.html)
Procedure Title	Marine gear cooler plumbing
Update Reason	Updated marine gear cooler graphics.
Date	17-Dec-2010
Section	Section 7 - Lubricating Oil System - Group 07
Procedure Number	28-007-025-tr (/qs3/pubsys2/xml/en/procedures/28/28-007-025-tr.html)
Procedure Title	Lubricating Oil Pan
Update Reason	Updated procedure reference in Preparatory and Finish sections with 008-137.
Date	17-Dec-2010
Section	Section 7 - Lubricating Oil System - Group 07
Procedure Number	28-007-027-tr (/qs3/pubsys2/xml/en/procedures/28/28-007-027-tr.html)
Procedure Title	Lubricating Oil Pan Adapter
Update Reason	Updated procedure reference in Preparatory and Finish sections with 008-137.
Date	16-Feb-2011
Section	Section 3 - Rocker Levers - Group 03
Procedure Number	28-003-006-tr (/qs3/pubsys2/xml/en/procedures/28/28-003-006-tr.html)
Procedure Title	Overhead Set (OBC)
Update Reason	Updated charts for valve locations to be adjusted.

Date	16-Feb-2011
Section	Section 7 - Lubricating Oil System - Group 07
Procedure Number	28-007-022-tr (/qs3/pubsys2/xml/en/procedures/28/28-007-022-tr.html)
Procedure Title	Lubricating Oil Jumper Cover
Update Reason	Procedure updated to include details of the different types of cover available. Also added details to the Remove and Install steps.
Date	18-Mar-2011
Section	Section 3 - Rocker Levers - Group 03
Procedure Number	28-003-011-tr (/qs3/pubsys2/xml/en/procedures/28/28-003-011-tr.html)
Procedure Title	Rocker Lever Cover
Update Reason	Updated procedure to show corrected graphics and content for removal on electronically actuated Injectors.
Date	31-Mar-2011
Section	Section 2 - Cylinder Head - Group 02
Procedure Number	28-002-004-tr (/qs3/pubsys2/xml/en/procedures/28/28-002-004-tr.html)
Procedure Title	Cylinder Head
Update Reason	Updated install section.
Date	09-May-2011
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	28-001-036 (/qs3/pubsys2/xml/en/procedures/28/28-001-036.html)
Procedure Title	Idler Gear, Camshaft
Update Reason	

Date	20-May-2011
Section	Section 8 - Cooling System - Group 08
Procedure Number	28-008-041-tr (/qs3/pubsys2/xml/en/procedures/28/28-008-041-tr.html)
Procedure Title	Marine gear oil cooler
Update Reason	Added to the outline. Remove by mistake.
Date	20-May-2011
Section	Section 8 - Cooling System - Group 08
Procedure Number	28-008-041-shopas (/qs3/pubsys2/xml/en/procedures/28/28-008-041-shopas.html)
Procedure Title	Marine gear oil cooler
Update Reason	Added to the outline. Remove by mistake.
Date	20-May-2011
Section	Section 8 - Cooling System - Group 08
Procedure Number	28-008-041-shopds (/qs3/pubsys2/xml/en/procedures/28/28-008-041-shopds.html)
Procedure Title	Marine gear oil cooler
Update Reason	Added to the outline. Remove by mistake.
Date	20-May-2011
Section	Section 8 - Cooling System - Group 08
Procedure Number	28-008-137-tr (/qs3/pubsys2/xml/en/procedures/28/28-008-137-tr.html)
Procedure Title	Marine gear cooler plumbing
Update Reason	Added to the outline. Remove by mistake.
Date	20-May-2011
Section	Section 8 - Cooling System - Group 08
Procedure Number	28-008-137-shopas (/qs3/pubsys2/xml/en/procedures/28/28-008-137-shopas.html)
Procedure Title	Marine gear cooler plumbing
Update Reason	Added to the outline. Remove by mistake.

Date	20-May-2011
Section	Section 8 - Cooling System - Group 08
Procedure Number	28-008-137-shopds (/qs3/pubsys2/xml/en/procedures/28/28-008-137-shopds.html)
Procedure Title	Marine gear cooler plumbing
Update Reason	Added to the outline. Remove by mistake.
Date	23-May-2011
Section	Section 6 - Injectors and Fuel Lines - Group 06
Procedure Number	28-006-025-tr (/qs3/pubsys2/xml/en/procedures/28/28-006-025-tr.html)
Procedure Title	Static Injection Timing
Update Reason	Updated procedure for clarity.
Date	01-Jun-2011
Section	Section 5 - Fuel System - Group 05
Procedure Number	28-005-016-tr (/qs3/pubsys2/xml/en/procedures/28/28-005-016-tr.html)
Procedure Title	Fuel Pump
Update Reason	Added General Information section.
Date	02-Jun-2011
Section	Section 3 - Rocker Levers - Group 03
Procedure Number	28-003-011-tr (/qs3/pubsys2/xml/en/procedures/28/28-003-011-tr.html)
Procedure Title	Rocker Lever Cover
Update Reason	Updated procedure to show corrected wiring harness bracket graphic.
Date	06-Jun-2011
Section	Section 8 - Cooling System - Group 08
Procedure Number	28-008-138 (/qs3/pubsys2/xml/en/procedures/28/28-008-138.html)
Procedure Title	Water Pump Plumbing
Update Reason	Added new procedure.

Date	17-Jun-2011
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	28-001-050-tr (/qs3/pubsys2/xml/en/procedures/28/28-001-050-tr.html)
Procedure Title	Vibration Damper, Double
Update Reason	Damper inspection criteria updated.
Date	17-Jun-2011
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	28-001-052-tr (/qs3/pubsys2/xml/en/procedures/28/28-001-052-tr.html)
Procedure Title	Vibration Damper, Viscous
Update Reason	Damper inspection criteria updated.
Date	11-Jul-2011
Section	Section V - Maintenance Specifications
Procedure Number	28-018-003-tr (/qs3/pubsys2/xml/en/procedures/28/28-018-003-tr.html)
Procedure Title	Lubricating Oil Recommendations and Specifications
Update Reason	Revised lubricating oil requirements
Date	19-Aug-2011
Section	Section 10 - Air Intake System - Group 10
Procedure Number	28-010-038-tr (/qs3/pubsys2/xml/en/procedures/28/28-010-038-tr.html)
Procedure Title	Turbocharger Axial Clearance
Update Reason	Removed from outline.
Date	25-Jul-2007
Section	Section 10 - Air Intake System - Group 10
Procedure Number	28-010-047-tr (/qs3/pubsys2/xml/en/procedures/28/28-010-047-tr.html)
Procedure Title	Turbocharger Radial Bearing Clearance
Update Reason	Removed from outline.

Date	22-Sep-2011
Section	Section 13 - Electrical Equipment - Group 13
Procedure Number	28-013-020-tr (/qs3/pubsys2/xml/en/procedures/28/28-013-020-tr.html)
Procedure Title	Starting Motor
Update Reason	Added a Note to the Remove and Install Steps to add spacers and gaskets.
Date	27-Sep-2011
Section	Section V - Maintenance Specifications
Procedure Number	28-018-015-tr (/qs3/pubsys2/xml/en/procedures/28/28-018-015-tr.html)
Procedure Title	General Engine
Update Reason	Updated cylinder call-outs for electronic 38 liter engines.
Date	27-Sep-2011
Section	Section 3 - Rocker Levers - Group 03
Procedure Number	28-003-006-tr (/qs3/pubsys2/xml/en/procedures/28/28-003-006-tr.html)
Procedure Title	Overhead Set (OBC)
Update Reason	Updated to show electronic engine cylinder numbering.
Date	12-Oct-2011
Section	Section 7 - Lubricating Oil System - Group 07
Procedure Number	28-007-031-tr (/qs3/pubsys2/xml/en/procedures/28/28-007-031-tr.html)
Procedure Title	Lubricating Oil Pump
Update Reason	Updated procedure to include the multiple types of pumps.

Date	10-Oct-2011
Section	Section 5 - Fuel System - Group 05
Procedure Number	28-005-016-tr (/qs3/pubsys2/xml/en/procedures/28/28-005-016-tr.html)
Procedure Title	Fuel Pump
Update Reason	Update procedure links in the Clean and Inspect for Reuse section.
Date	25-Oct-2011
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021528-t108-tr (/qs3/pubsys2/xml/en/manual/4021528/4021528-t108-tr.html)
Procedure Title	Lubricating or Transmission Oil in the Coolant
Update Reason	Revised turbocharger bearing housing is cracked or porous step.
Date	04-Nov-2011
Section	Section 7 - Lubricating Oil System - Group 07
Procedure Number	28-007-021-tr (/qs3/pubsys2/xml/en/procedures/28/28-007-021-tr.html)
Procedure Title	Lubricating Oil High Pressure Relief Valve
Update Reason	Updated procedure to include new prep step, removed Clean and Inspect for Reuse section, and added necessary engine references.
Date	07-Nov-2011
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	28-001-006-tr (/qs3/pubsys2/xml/en/procedures/28/28-001-006-tr.html)
Procedure Title	Bearings, Main
Update Reason	Revise reference for crankshaft end clearance specification.

Date	07-Nov-2011
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	28-001-007-tr (/qs3/pubsys2/xml/en/procedures/28/28-001-007-tr.html)
Procedure Title	Bearing, Thrust
Update Reason	Added end clearance measurement techniques once engine is installed into an application.

Date	14-Nov-2011
Section	Section 5 - Fuel System - Group 05
Procedure Number	28-005-016-tr (/qs3/pubsys2/xml/en/procedures/28/28-005-016-tr.html)
Procedure Title	Fuel Pump
Update Reason	Revised fuel rail pressure sensor torque value and added note.

Date	23-Nov-2011
Section	Section 8 - Cooling System - Group 08
Procedure Number	28-008-057-tr (/qs3/pubsys2/xml/en/procedures/28/28-008-057-tr.html)
Procedure Title	Sea Water Pump
Update Reason	Added Remove and Install instructions for Jabsco™ sea water pumps.

Date	05-Dec-2011
Section	Section 08 - Cooling System
Procedure Number	28-008-062-tr (/qs3/pubsys2/xml/en/procedures/28/28-008-062-tr.html)
Procedure Title	Water Pump
Update Reason	Assemble Step, revised text to install the oil seal with the stepped lip facing down towards the impeller.

Date	13-Dec-2011
Section	Section V - Maintenance Specifications
Procedure Number	00-018-004 (/qs3/pubsys2/xml/en/procedures/00/00-018-004.html)
Procedure Title	Coolant Recommendations and Specifications
Update Reason	Updated freezing point chart with the addition of Celsius and Fahrenheit values.
Date	19-Dec-2011
Section	Section 1 - Cylinder Block
Procedure Number	28-001-031 (/qs3/pubsys2/xml/en/procedures/28/28-001-031.html)
Procedure Title	Gear Cover, Front
Update Reason	Added tr effectivity to Preparatory Steps and Finishing Steps
Date	19-Dec-2011
Section	Section 3 - Rocker Levers
Procedure Number	28-003-006-tr (/qs3/pubsys2/xml/en/procedures/28/28-003-006-tr.html)
Procedure Title	Overhead Set (OBC)
Update Reason	Added: Use service tool Part Number 3163171 (intake) or 3163172 (exhaust).
Date	19-Dec-2011
Section	Section 3 - Rocker Levers
Procedure Number	28-003-007-tr (/qs3/pubsys2/xml/en/procedures/28/28-003-007-tr.html)
Procedure Title	Overhead Set (Travel Method)
Update Reason	Added: Use service tool Part Number 3163171 (intake) or 3163172 (exhaust).

Date	19-Jan-2012
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021528-t057-tr (/qs3/pubsys2/xml/en/manual/4021528/4021528-t057-tr.html)
Procedure Title	This troubleshooting symptom tree has been updated to include details of inspecting the filter screen in the fuel pump shutoff valve block.
Update Reason	
Date	19-Jan-2012
Section	Section 5 - Fuel System - Group 05
Procedure Number	28-005-016-tr (/qs3/pubsys2/xml/en/procedures/28/28-005-016-tr.html)
Procedure Title	Fuel Pump
Update Reason	This procedure has been updated to include details of inspecting the filter screen in the fuel pump shutoff valve block..
Date	20-Feb-2012
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021528-t021-tr (/qs3/pubsys2/xml/en/manual/4021528/4021528-t021-tr.html)
Procedure Title	Coolant Loss - Internal
Update Reason	Add node: Transmission oil cooler or torque converter cooler is leaking.
Date	08-Mar-2012
Section	Section 3 - Rocker Levers
Procedure Number	28-003-013-tr (/qs3/pubsys2/xml/en/procedures/28/28-003-013-tr.html)
Procedure Title	Rocker Lever Housing
Update Reason	Revised Remove Step, adding the use of the Water Tube Driver, ST-1319.

Date	05-Jul-2007
Section	Section 7 - Lubricating Oil System - Group 07
Procedure Number	28-007-013-tr (/qs3/pubsys2/xml/en/procedures/28/28-007-013-tr.html)
Procedure Title	Lubricating Oil Filter (Spin-On)
Update Reason	Pressure Differential Test has been created and added to procedure.
Date	30-Apr-2012
Section	Section 5 - Fuel System - Group 05
Procedure Number	28-005-016-tr (/qs3/pubsys2/xml/en/procedures/28/28-005-016-tr.html)
Procedure Title	Fuel Pump
Update Reason	Added information to General Information section about relieving rail pressure before removing parts from the fuel pump.
Date	02-May-2012
Section	Section 2 - Cylinder Head - Group 02
Procedure Number	28-002-004-tr (/qs3/pubsys2/xml/en/procedures/28/28-002-004-tr.html)
Procedure Title	Cylinder Head
Update Reason	Added new valve spring technical information.
Date	23-May-2012
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	28-001-043-tr (/qs3/pubsys2/xml/en/procedures/28/28-001-043-tr.html)
Procedure Title	Piston
Update Reason	Added new General Information Step concerning the new piston and second ring release for the QSK50.

Date	23-May-2012
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	28-001-047-tr (/qs3/pubsys2/xml/en/procedures/28/28-001-047-tr.html)
Procedure Title	Piston Rings
Update Reason	Added new second ring gap information per the new piston and second ring release for the QSK50.

Date	23-May-2012
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	28-001-016-tr (/qs3/pubsys2/xml/en/procedures/28/28-001-016-tr.html)
Procedure Title	Crankshaft
Update Reason	Updated procedure by adding information to clarify the use of Scotch-Brite™ pads or crocus cloth in the Clean and Inspect section.

Date	23-May-2012
Section	Section V - Specifications
Procedure Number	28-018-016-tr (/qs3/pubsys2/xml/en/procedures/28/28-018-016-tr.html)
Procedure Title	Fuel System
Update Reason	Updated pressure specifications by providing pressure values in kPa and in-Hg units of measurement.

Date	30-May-2012
Section	Section 6 - Injectors and Fuel Lines - Group 06
Procedure Number	28-006-017-tr (/qs3/pubsys2/xml/en/procedures/28/28-006-017-tr.html)
Procedure Title	Fuel Filter Head
Update Reason	Updated graphics to show fittings for new fuel lines.

Date	30-May-2012
Section	Section 6 - Injectors and Fuel Lines - Group 06
Procedure Number	28-006-024-tr (/qs3/pubsys2/xml/en/procedures/28/28-006-024-tr.html)
Procedure Title	Fuel Supply Lines
Update Reason	Updated graphics to show new fuel supply lines.

Date	08-Jun-2012
Section	Section 6 - Injectors and Fuel Lines - Group 06
Procedure Number	28-006-020-tr (/qs3/pubsys2/xml/en/procedures/28/28-006-020-tr.html)
Procedure Title	Fuel Inlet Restriction
Update Reason	Updated descriptions for clarity and updated pressure specifications by providing pressure values in kPa and in-Hg units of measurement.

Date	27-Jun-2012
Section	Section 6 - Injectors and Fuel Lines
Procedure Number	28-006-025-tr (/qs3/pubsys2/xml/en/procedures/28/28-006-025-tr.html)
Procedure Title	Static Injection Timing
Update Reason	Changed Control Part List (CPL) Manual, Bulletin to 3379133, added refer to Procedure 99-850-013, and removed Timing Code Tables.

Date	05-Jul-2012
Section	Section 3 - Rocker Levers
Procedure Number	28-003-013-tr (/qs3/pubsys2/xml/en/procedures/28/28-003-013-tr.html)
Procedure Title	Rocker Lever Housing
Update Reason	Revised Install step to include information to prevent oil leaks from the rocker housing joint.

Date	25-Jun-2012
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	28-001-028-tr (/qs3/pubsys2/xml/en/procedures/28/28-001-028-tr.html)
Procedure Title	Cylinder Liner
Update Reason	Updated the Remove step to add instruction that liner seals must be discarded if removed from the liner.

Date	03-Aug-2012
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	28-001-016-tr (/qs3/pubsys2/xml/en/procedures/28/28-001-016-tr.html)
Procedure Title	Crankshaft
Update Reason	Added reuse criteria for crankshaft front nose.

Date	03-Aug-2012
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	28-001-023-tr (/qs3/pubsys2/xml/en/procedures/28/28-001-023-tr.html)
Procedure Title	Crankshaft Seal, Front
Update Reason	Revised entire procedure emphasis on the seal and associated tooling.

Date	20-Aug-2012
Section	Section TS - Troubleshooting System
Procedure Number	4021528-t102-tr (/qs3/pubsys2/xml/en/manual/4021528/4021528-t102-tr.html)
Procedure Title	Lubricating Oil Consumption Excessive
Update Reason	Changed first step to include inspection of OEM components for external oil leaks.

Date	20-Aug-2012
Section	Section 8 - Cooling System - Group 08
Procedure Number	28-022-001_08-tr (/qs3/pubsys2/xml/en/procedures/28/28-022-001_08-tr.html)
Procedure Title	Service Tools
Update Reason	Update service tool list.
Date	20-Aug-2012
Section	Section 8 - Cooling System - Group 08
Procedure Number	28-008-062-tr (/qs3/pubsys2/xml/en/procedures/28/28-008-062-tr.html)
Procedure Title	Water Pump
Update Reason	Update service tool part numbers.
Date	17-Aug-2012
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	28-001-005-tr (/qs3/pubsys2/xml/en/procedures/28/28-001-005-tr.html)
Procedure Title	Bearings, Connecting Rod
Update Reason	Updated procedure, adding service tool Part Number 3822764.
Date	17-Aug-2012
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	28-001-054-tr (/qs3/pubsys2/xml/en/procedures/28/28-001-054-tr.html)
Procedure Title	Piston and Connecting Rod Assembly
Update Reason	Updated procedure, adding service tool Part Number 3822764.

Date	17-Aug-2012
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	28-001-022_001-tr (/qs3/pubsys2/xml/en/procedures/28/28-001-022_001-tr.html)
Procedure Title	Bearings, Connecting Rod
Update Reason	Updated procedure, adding service tool Part Number 3822764.

Date	24-Aug-2012
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	28-001-063-tr (/qs3/pubsys2/xml/en/procedures/28/28-001-063-tr.html)
Procedure Title	Idler Shaft, Water Pump Idler Gear
Update Reason	Added procedure to Section 1 in outline.

Date	04-Sep-2012
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	28-001-026-tr (/qs3/pubsys2/xml/en/procedures/28/28-001-026-tr.html)
Procedure Title	Cylinder Block
Update Reason	Procedure has been revised and updated with correct information about specifications of lower counterbore diameter is valid for specific block design to be used and sealant part number.

Date	04-Sep-2012
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	28-001-027 (/qs3/pubsys2/xml/en/procedures/28/28-001-027.html)
Procedure Title	Cylinder Block and Liner Seats
Update Reason	Procedure has been revised and updated with correct information about specifications of lower counterbore diameter is valid for specific block design to be used and sealant part number.

Date	04-Sep-2012
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	28-001-028-tr (/qs3/pubsys2/xml/en/procedures/28/28-001-028-tr.html)
Procedure Title	Cylinder Liner
Update Reason	Liner installation step has been revised and updated with correct information about specifications to be used and sealant part number.
Date	04-Sep-2012
Section	Section 8 - Cooling System - Group 08
Procedure Number	28-008-062-tr (/qs3/pubsys2/xml/en/procedures/28/28-008-062-tr.html)
Procedure Title	Water Pump
Update Reason	Added water pump weep hole tube and filter information.
Date	24-Sep-2012
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	28-001-033-tr (/qs3/pubsys2/xml/en/procedures/28/28-001-033-tr.html)
Procedure Title	Gear Housing, Front
Update Reason	Revised procedure links in Preparatory and Finishing sections.
Date	24-Oct-2012
Section	Section 3 - Rocker Levers - Group 03
Procedure Number	28-003-011-tr (/qs3/pubsys2/xml/en/procedures/28/28-003-011-tr.html)
Procedure Title	Rocker Lever Cover
Update Reason	In the electronically actuated injector section, a statement is added to discard the o-ring after removing it from the rocker lever cover plate.

Date	28-Nov-2012
Section	Section 8 - Cooling System - Group 08
Procedure Number	28-008-062-tr (/qs3/pubsys2/xml/en/procedures/28/28-008-062-tr.html)
Procedure Title	Water Pump
Update Reason	Updated procedure on oil seal installation instructions to account for the new seal and difficulties installing it.
Date	06-Jan-2013
Section	Section 6 - Injectors
Procedure Number	28-022-001_06 (/qs3/pubsys2/xml/en/procedures/28/28-022-001_06.html)
Procedure Title	Fuel System Tester
Update Reason	Added Service Tool Part Number 4918612.
Date	25-Jan-2013
Section	Section 10 - Air Intake System - Group 10
Procedure Number	28-010-002-tr (/qs3/pubsys2/xml/en/procedures/28/28-010-002-tr.html)
Procedure Title	Aftercooler Assembly
Update Reason	Added Pressure Test to procedure.
Date	31-Jan-2013
Section	Section 8 - Cooling System - Group 08
Procedure Number	28-008-015-tr (/qs3/pubsys2/xml/en/procedures/28/28-008-015-tr.html)
Procedure Title	Coolant Thermostat Housing Support
Update Reason	The procedure was revised and two additional tasks and torque specification have been added in the Install Step.

Date	18-Jan-2013
Section	Section DS - Engine Disassembly - Group 00
Procedure Number	28-006-038-shopds (/qs3/pubsys2/xml/en/procedures/28/28-006-038-shopds.html)
Procedure Title	STC Oil Manifold
Update Reason	Correction to step timing control (STC) oil manifold capscrew torque value.

Date	18-Jan-2013
Section	Section AS - Engine Assembly - Group 00
Procedure Number	28-006-038-shopas (/qs3/pubsys2/xml/en/procedures/28/28-006-038-shopas.html)
Procedure Title	STC Oil Manifold
Update Reason	Correction to step timing control (STC) oil manifold capscrew torque value.

Date	21-Apr-2013
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021528-t092 (/qs3/pubsys2/xml/en/manual/4021528/4021528-t092.html)
Procedure Title	Fuel in the Lubricating Oil
Update Reason	Updated "Fuel pump seal is leaking" and Injector is malfunctioning" steps.

Date	22-Apr-2013
Section	Section 5 - Fuel System - Group 05
Procedure Number	28-005-016-tr (/qs3/pubsys2/xml/en/procedures/28/28-005-016-tr.html)
Procedure Title	Fuel Pump
Update Reason	Added Flush section to procedure.

Date	10-May-2013
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	28-001-043-tr (/qs3/pubsys2/xml/en/procedures/28/28-001-043-tr.html)
Procedure Title	Piston
Update Reason	New General Information section was added to distinguish between the two types of piston available for KV tier 1 engines.
Date	28-June-2013
Section	Section 3 - Rocker Levers - Group 3
Procedure Number	28-003-013-tr (/qs3/pubsys2/xml/en/procedures/28/28-003-013-tr.html)
Procedure Title	Rocker Lever Housing
Update Reason	Install Step: Added statement, checking for capscrews for damage. Change capscrew torque to 122 N.m [90 ft-lb].
Date	29-Jun-2013
Section	Section 8 - Cooling System - Group 08
Procedure Number	28-008-062-tr (/qs3/pubsys2/xml/en/procedures/28/28-008-062-tr.html)
Procedure Title	Water Pump
Update Reason	Refer the Tool Catalog, Publication Number 3824836 to determine which tool number to use depending on the water pump seal part number.
Date	06-Aug-2013
Section	Section 6 - Injectors and Fuel Lines - Group 06
Procedure Number	28-006-020-tr (/qs3/pubsys2/xml/en/procedures/28/28-006-020-tr.html)
Procedure Title	Fuel Inlet Restriction
Update Reason	Revised the Maximum Fuel Inlet Restriction values for Stage 1 and Stage 2 filters.

Date	06-Aug-2013
Section	Section V - Maintenance Guidelines
Procedure Number	28-018-024-tr (/qs3/pubsys2/xml/en/procedures/28/28-018-024-tr.html)
Procedure Title	Cummins®/Fleetguard® Filter Specifications
Update Reason	Updated filter specifications and part numbers.
Date	05-Aug-2013
Section	Section TT - Troubleshooting Performance Troubleshooting Tree - New Format)
Procedure Number	05-t02-1001 (/qs3/pubsys2/xml/en/procedures/05/05-t02-1001.html)
Procedure Title	Engine Performance Troubleshooting Tree
Update Reason	Guided Step 3H: Added, Repair fuel rail or leaking component to YES.
Date	13-Aug-2013
Section	Section 6 - Injectors and Fuel Lines - Group 06
Procedure Number	28-006-075-tr (/qs3/pubsys2/xml/en/procedures/28/28-006-075-tr.html)
Procedure Title	Fuel Filter (Stage 1)
Update Reason	Added information and illustration for FH239 fuel filter.
Date	16-Aug-2013
Section	Section 8 - Cooling System - Group 08
Procedure Number	28-008-109-tr (/qs3/pubsys2/xml/en/procedures/28/28-008-109-tr.html)
Procedure Title	Low-Temperature Aftercooler (LTA) Water Pump
Update Reason	Updated procedure due to SQP number DAV10001 requiring an oil fed bearing.

Date	26-Aug-2013
Section	Section 14 - Engine Testing - Group 14
Procedure Number	00-014-012 (/qs3/pubsys2/xml/en/procedures/00/00-014-012.html)
Procedure Title	Engine Compression
Update Reason	Updated Preparatory Step and Finishing Step procedure links.
Date	28-Aug-2013
Section	Section V - Maintenance Specifications
Procedure Number	102-018-002 (/qs3/pubsys2/xml/en/procedures/102/102-018-002.html)
Procedure Title	Fuel Recommendations and Specifications
Update Reason	Updated fuel specifications.
Date	28-Aug-2013
Section	Section V - Maintenance Guidelines
Procedure Number	20-018-016 (/qs3/pubsys2/xml/en/procedures/20/20-018-016.html)
Procedure Title	Fuel System
Update Reason	Updated fuel inlet maximum restriction.
Date	16-Sep-2013
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	28-001-005-tr (/qs3/pubsys2/xml/en/procedures/28/28-001-005-tr.html)
Procedure Title	Bearings, Connecting Rod
Update Reason	Added the required step of loosening the capscrews after first stage of tightening.

Date	16-Sep-2013
Section	Section 3 - Rocker Levers - Group 03
Procedure Number	28-003-009-tr (/qs3/pubsys2/xml/en/procedures/28/28-003-009-tr.html)
Procedure Title	Rocker Lever Assembly
Update Reason	The procedure has been revised by adding the Clean and Inspect for Reuse an additional information.

Date	06-Nov-2013
Section	Section 16 - Mounting Adaptations - Group 16
Procedure Number	28-016-002 (/qs3/pubsys2/xml/en/procedures/28/28-016-002.html)
Procedure Title	Engine Support Bracket, Front
Update Reason	Added reference to the original equipment manufacturer service manual in the Install Step.

Date	05-Nov-2013
Section	Section 16 - Mounting Adaptations - Group 16
Procedure Number	102-016-008 (/qs3/pubsys2/xml/en/procedures/102/102-016-008.html)
Procedure Title	Flywheel Ring Gear
Update Reason	Updated procedure for improved clarity and safety.

Date	16-Dec-2013
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	28-001-005-tr (/qs3/pubsys2/xml/en/procedures/28/28-001-005-tr.html)
Procedure Title	Bearings, Connecting Rod
Update Reason	Added information about visual inspection of the bearings in the Clean and Inspect for Reuse step.

Date	16-Dec-2013
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	28-001-006-tr (/qs3/pubsys2/xml/en/procedures/28/28-001-006-tr.html)
Procedure Title	Bearings, Main
Update Reason	Added information about visual inspection of the bearings in the Clean and Inspect for Reuse step.
Date	17-Dec-2013
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	28-001-058-tr (/qs3/pubsys2/xml/en/procedures/28/28-001-058-tr.html)
Procedure Title	Cylinder Block Counterbore
Update Reason	Information about repair sleeve reference dimensions in the REPAIR section has been added to the procedure contents.
Date	08-Jan-2014
Section	Section 6 - Injectors and Fuel Lines - Group 06
Procedure Number	28-006-020-tr (/qs3/pubsys2/xml/en/procedures/28/28-006-020-tr.html)
Procedure Title	Fuel Inlet Restriction
Update Reason	Correction to the Initial Check section.
Date	14-Jan-2014
Section	Section TS - Troubleshooting System
Procedure Number	4021528-t102-tr (/qs3/pubsys2/xml/en/manual/4021528/4021528-t102-tr.html)
Procedure Title	Lubricating Oil Consumption Excessive
Update Reason	Added step three to TS tree. Verifying original equipment manufacturer (OEM) lubricating oil system components.

Date	17-Feb-2014
Section	Section 2 - Cylinder Head - Group 02
Procedure Number	28-002-004-tr (/qs3/pubsys2/xml/en/procedures/28/28-002-004-tr.html)
Procedure Title	Cylinder Head
Update Reason	Updated valve lapping processes under Grind and Machine steps.
Date	17-Mar-2014
Section	Section 6 - Injectors and Fuel Lines - Group 06
Procedure Number	102-006-061 (/qs3/pubsys2/xml/en/procedures/102/102-006-061.html)
Procedure Title	Fuel Pressure Relief Valve
Update Reason	Added the Fuel Pressure Valve Return Flow Test
Date	17-Jun-2014
Section	Section 8 - Cooling System - Group 08
Procedure Number	28-008-035-tr (/qs3/pubsys2/xml/en/procedures/28/28-008-035-tr.html)
Procedure Title	Fan Hub Bracket
Update Reason	Procedure updated to include installation instructions for the newly released fan hub bracket.
Date	17-Jun-2014
Section	Section 8 - Cooling System - Group 08
Procedure Number	28-008-036-tr (/qs3/pubsys2/xml/en/procedures/28/28-008-036-tr.html)
Procedure Title	Fan Hub, Belt Driven
Update Reason	Procedure updated to include installation instructions for the newly released fan hub bracket.

Date	04-Sep-2014
Section	Section 3 Rocker Levers - Group 03
Procedure Number	28-003-011-tr (/qs3/pubsys2/xml/en/procedures/28/28-003-011-tr.html)
Procedure Title	Rocker Lever Cover
Update Reason	Added instructions to pry evenly on the rocker lever cover.
Date	08-Oct-2014
Section	Section TS - Troubleshooting Symptoms - Group t00
Procedure Number	4021528-t107-tr (/qs3/pubsys2/xml/en/manual/4021528/4021528-t107-tr.html)
Procedure Title	Lubricating Oil Temperature Above Specification
Update Reason	Updated TS tree with correct wording for field service on lubricating oil above or below specification.
Date	10-Sep-2014
Section	Section F - Familiarization
Procedure Number	28-200-002-tr (/qs3/pubsys2/xml/en/procedures/28/28-200-002-tr.html)
Procedure Title	Flow Diagram, Lubricating Oil System
Update Reason	Revised oil flow to and through the fuel pump accessory drive.
Date	16-Sep-2014
Section	Section 9 - Drive Units
Procedure Number	28-009-011 (/qs3/pubsys2/xml/en/procedures/28/28-009-011.html)
Procedure Title	Fuel Pump Drive
Update Reason	Accessory Drive Bushing Trust Face Distance and Bushing Inside Diameter revised.

Date	06-Nov-2014
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	28-001-024-tr (/qs3/pubsys2/xml/en/procedures/28/28-001-024-tr.html)
Procedure Title	Crankshaft Seal, Rear
Update Reason	Added information concerning Service Tool, Part Number 3824424 to the General Information Step and the Install Step.
Date	06-Nov-2014
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	28-022-001_01-tr (/qs3/pubsys2/xml/en/procedures/28/28-022-001_01-tr.html)
Procedure Title	Service Tools
Update Reason	Added Service Tool, Part Number 3824424.
Date	12-Nov-2014
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	28-001-026-tr (/qs3/pubsys2/xml/en/procedures/28/28-001-026-tr.html)
Procedure Title	Cylinder Block
Update Reason	Removed the information concerning Service Tool, Part Number 3824424 to the General Information Step.
Date	13-Nov-2014
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	28-001-028-tr (/qs3/pubsys2/xml/en/procedures/28/28-001-028-tr.html)
Procedure Title	Cylinder Liner
Update Reason	Removed note about unavailability of oversize liners for engines with electronically actuated injectors (MCRS) which became obsolete after publishing TSB140090 announcing the release of such cylinder liners.

Date	20-Nov-2014
Section	Section 9 - Drive Units - Group 09
Procedure Number	28-009-031-tr (/qs3/pubsys2/xml/en/procedures/28/28-009-031-tr.html)
Procedure Title	Water Pump Drive Gear and Shaft
Update Reason	Improved Assemble steps and added new Instal steps.

Date	20-Nov-2014
Section	Section 8 - Cooling System
Procedure Number	28-008-111 (/qs3/pubsys2/xml/en/procedures/28/28-008-111.html)
Procedure Title	Pulley, Fan Idler
Update Reason	Updated recommendation for using currently available greases.

Date	20-Nov-2014
Section	Section 8 - Cooling System
Procedure Number	28-008-036-tr (/qs3/pubsys2/xml/en/procedures/28/28-008-036-tr.html)
Procedure Title	Pulley, Fan Idler
Update Reason	Updated recommendation for using currently available greases.

Date	06-Jan-2015
Section	Section 6 - Injectors and Fuel Lines - Group 06
Procedure Number	28-006-026-tr (/qs3/pubsys2/xml/en/procedures/28/28-006-026-tr.html)
Procedure Title	Injector
Update Reason	Corrected the injector hold-down clamp orientation for installing the injector.

Date	8-Jan-2015
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	28-001-005-tr (/qs3/pubsys2/xml/en/procedures/28/28-001-005-tr.html)
Procedure Title	Bearings, Connecting Rod
Update Reason	Changed the values of axial clearance of connecting rods to provide consistent service information.

Date	8-Jan-2015
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	28-001-054-tr (/qs3/pubsys2/xml/en/procedures/28/28-001-054-tr.html)
Procedure Title	Piston and Connecting Rod Assembly
Update Reason	Changed the values of axial clearance of connecting rods to provide consistent service information.

Date	01-Apr-2015
Section	Section 16 - Mounting Adaptations - Group 16
Procedure Number	28-016-002-tr (/qs3/pubsys2/xml/en/procedures/28/28-016-002-tr.html)
Procedure Title	Engine Support Bracket, Front
Update Reason	Procedure has been updated with currently available greases recommendations.

Date	08-Apr-2015
Section	Section 8 - Cooling System - Group 08
Procedure Number	28-008-018-tr (/qs3/pubsys2/xml/en/procedures/28/28-008-018-tr.html)
Procedure Title	Cooling System
Update Reason	Added directions to allow the engine to cool before draining coolant.

Date	07-May-2015
Section	Section 8 - Cooling System
Procedure Number	28-008-036-tr (/qs3/pubsys2/xml/en/procedures/28/28-008-036-tr.html)
Procedure Title	Fan Hub, Belt Driven
Update Reason	Updated grease(s) recommendations.
Date	07-May-2015
Section	Section 8 - Cooling System
Procedure Number	28-008-111 (/qs3/pubsys2/xml/en/procedures/28/28-008-111.html)
Procedure Title	Pulley, Fan Idler
Update Reason	Updated grease(s) recommendations.
Date	19-May-2015
Section	Section 14 - Engine Testing - Group 14
Procedure Number	28-014-002-tr (/qs3/pubsys2/xml/en/procedures/28/28-014-002-tr.html)
Procedure Title	Engine Testing (Chassis Dynamometer)
Update Reason	Changed task from Intake Manifold Air Temperature Control - Chassis Dynamometer test to Exhaust Air Restriction.
Date	06-Jun-2015
Section	Section 3 - Rocker Levers - Group 03
Procedure Number	28-003-006-tr (/qs3/pubsys2/xml/en/procedures/28/28-003-006-tr.html)
Procedure Title	Overhead Set (OBC)
Update Reason	Corrected the torque spec's on the Adjust Section to the correct specification is 0.68 n.m [6 in-lb].

Date	15-Jul-2015
Section	Section 16 - Mounting Adaptations - Group 16
Procedure Number	28-016-006-tr (/qs3/pubsys2/xml/en/procedures/28/28-016-006-tr.html)
Procedure Title	Flywheel Housing
Update Reason	Updated torque values in redowel section.
Date	04-Aug-2015
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	28-001-028-tr (/qs3/pubsys2/xml/en/procedures/28/28-001-028-tr.html)
Procedure Title	Cylinder Liner
Update Reason	Matched the order of the steps in the procedure 28-001-028-shopas with the order of the steps in main procedure.
Date	25-Aug-2015
Section	Section 1 - Cylinder Block
Procedure Number	28-001-016-tr (/qs3/pubsys2/xml/en/procedures/28/28-001-016-tr.html)
Procedure Title	Camshaft
Update Reason	Revised Crankshaft Main Bearing Journal and Crankshaft Connecting Rod Journal Outside Diameter tables.
Date	14-Sep-2015
Section	Section 16 - Mounting Adaptations - Group 16
Procedure Number	28-016-006-tr (/qs3/pubsys2/xml/en/procedures/28/28-016-006-tr.html)
Procedure Title	Flywheel Housing
Update Reason	Specified all capscrews need to be Grade 8.

Date	22-Sep-2015
Section	Section 9 - Drive Units - Group 09
Procedure Number	28-009-072 (/qs3/pubsys2/xml/en/procedures/28/28-009-072.html)
Procedure Title	Air Compressor Drive Gear
Update Reason	Added new procedure to outline.
Date	16-Oct-2015
Section	Section 6 - Injectors and Fuel Lines - Group 06
Procedure Number	28-006-025-tr (/qs3/pubsys2/xml/en/procedures/28/28-006-025-tr.html)
Procedure Title	Static Injection Timing
Update Reason	Replaced the graphics at the end of the Measure section with text and tables.
Date	27-Oct-2015
Section	Section 14 - Engine Testing - Group 14
Procedure Number	28-014-008 (/qs3/pubsys2/xml/en/procedures/28/28-014-008.html)
Procedure Title	Engine Testing (In Chassis)
Update Reason	Added procedure to manual.
Date	30-Oct-2015
Section	Section 16 - Mounting Adaptations - Group 16
Procedure Number	28-016-006-tr (/qs3/pubsys2/xml/en/procedures/28/28-016-006-tr.html)
Procedure Title	Flywheel Housing
Update Reason	Updated Remove and Install sections to reflect Flywheel Housing without Rear Seal and Flywheel Housing with Rear Seal.

Date	30-Oct-2015
Section	Section 7 - Lubricating Oil System - Group 07
Procedure Number	28-007-025-tr (/qs3/pubsys2/xml/en/procedures/28/28-007-025-tr.html)
Procedure Title	Lubricating Oil Pan
Update Reason	Updated lubricating oil drain plug torque value.
Date	22-Jan-2016
Section	Section 6 - Injectors and Fuel Lines - Group 06
Procedure Number	28-006-051-tr (/qs3/pubsys2/xml/en/procedures/28/28-006-051-tr.html)
Procedure Title	Injector Supply Lines (High Pressure)
Update Reason	Updated pressure relief warning in Leak Test section for C3 fuel systems.
Date	01-Feb-2016
Section	Section TT - Troubleshooting Symptoms (New Format)
Procedure Number	28-t02-1005a (/qs3/pubsys2/xml/en/procedures/28/28-t02-1005a.html)
Procedure Title	Vibration Excessive
Update Reason	Added Guided Steps 1J and 1K.
Date	17-Mar-2016
Section	Section 9 - Drive Units - Group 09
Procedure Number	28-009-010-tr (/qs3/pubsys2/xml/en/procedures/28/28-009-010-tr.html)
Procedure Title	Alternator Drive Pulley
Update Reason	Updated pulley reuse criteria.

Date	29-Mar-2016
Section	Section 7 - Lubricating Oil System
Procedure Number	28-007-031-tr (/qs3/pubsys2/xml/en/procedures/28/28-007-031-tr.html)
Procedure Title	Lubricating Oil Pump
Update Reason	Added information to visually check the gear cover for corner radius, and added slow speed pump information.

Date	27-Apr-2016
Section	Section 16 - Mounting Adaptations - Group 16
Procedure Number	28-016-005-tr (/qs3/pubsys2/xml/en/procedures/28/28-016-005-tr.html)
Procedure Title	Flywheel
Update Reason	Corrected the calculation to determine flywheel axial runout.

Date	11-May-2016
Section	Section 08 - Cooling System
Procedure Number	28-008-062-tr (/qs3/pubsys2/xml/en/procedures/28/28-008-062-tr.html)
Procedure Title	Water Pump
Update Reason	Corrected the torque specification for the water pump mounting capscrews .

Date	07-Jun-2016
Section	Section 8 - Cooling System - Group 08
Procedure Number	28-008-036-tr (/qs3/pubsys2/xml/en/procedures/28/28-008-036-tr.html)
Procedure Title	Fan Hub, Belt Driven
Update Reason	Procedure updated to add limits for the fan hub shaft in order to be used for rebuild.

Date	08-Jun-2016
Section	Section 7 - Lubricating Oil System - Group 07
Procedure Number	28-007-007-tr (/qs3/pubsys2/xml/en/procedures/28/28-007-007-tr.html)
Procedure Title	Lubricating Oil Cooler Element
Update Reason	Changes have been made to the cleaning recommendations for oil cooler elements.
Date	19-Aug-2016
Section	Section 6 - Injectors and Fuel Lines - Group 06
Procedure Number	28-006-017-tr (/qs3/pubsys2/xml/en/procedures/28/28-006-017-tr.html)
Procedure Title	Fuel Filter Head
Update Reason	Updated to include Front Mounted with Electronically Actuated Injector.
Date	28-Sep-2016
Section	Section 7 - Lubricating Oil System - Group 07
Procedure Number	102-007-083 (/qs3/pubsys2/xml/en/procedures/102/102-007-083.html)
Procedure Title	Lubricating Oil and Filter Analysis
Update Reason	Added to outline.
Date	07-Oct-2016
Section	Section 4 - Cam Followers/Tappets
Procedure Number	28-004-002-tr (/qs3/pubsys2/xml/en/procedures/28/28-004-002-tr.html)
Procedure Title	Cam Follower Cover
Update Reason	Added to Preparatory/Finishing Step: Remove the left and right bank injector and exhaust gas temperature sensor harness.

Date	31-Oct-2016
Section	Section 3 - Rocker Levers
Procedure Number	28-003-011-tr (/qs3/pubsys2/xml/en/procedures/28/28-003-011-tr.html)
Procedure Title	Rocker Lever Cover
Update Reason	Added: Remove the injector (EGTS harness.
Date	03-Jan-2017
Section	Section 1 - Cylinder Block
Procedure Number	28-001-010 (/qs3/pubsys2/xml/en/procedures/28/28-001-010.html)
Procedure Title	Camshaft Bushings
Update Reason	Revised the Camshaft Bushing Bore in the Block table.
Date	30-Jan-2017
Section	Section 1 - Cylinder Block
Procedure Number	28-001-006-tr (/qs3/pubsys2/xml/en/procedures/28/28-001-006-tr.html)
Procedure Title	Bearings, Main
Update Reason	Removed statement about measuring the shine area of the thrust bearing.
Date	06-Feb-2017
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	28-001-024-tr (/qs3/pubsys2/xml/en/procedures/28/28-001-024-tr.html)
Procedure Title	Crankshaft Seal, Rear
Update Reason	Total internal runout measurements for the rear seal have been added to the Install Step.

Date	21-Feb-2017
Section	Section 1 - Cylinder Block
Procedure Number	28-001-027-tr (/qs3/pubsys2/xml/en/procedures/28/28-001-027-tr.html)
Procedure Title	Cylinder Block and Liner Seats
Update Reason	Updated Oversize tables.
Date	22-Feb-2017
Section	Section 1 - Cylinder Block
Procedure Number	28-001-026-tr (/qs3/pubsys2/xml/en/procedures/28/28-001-026-tr.html)
Procedure Title	Cylinder Block
Update Reason	Updated Oversize table.
Date	02-Mar-2017
Section	Section 14 - Engine Testing - Group 14
Procedure Number	99-014-024 (/qs3/pubsys2/xml/en/procedures/99/99-014-024.html)
Procedure Title	Fluorescent Tracer Dye Test
Update Reason	Added procedure to manual.
Date	13-Mar-2017
Section	Section 1 - Cylinder Block
Procedure Number	20-010-016-tr (/qs3/pubsys2/xml/en/procedures/20/20-010-016-tr.html)
Procedure Title	Crankshaft
Update Reason	Added, Remove/Install the cylinder head and piston and connecting rod assembly to Preparatory and Finishing Step.

Date	27-Mar-2017
Section	Section 8 - Cooling System - Group 08
Procedure Number	28-008-002-tr (/qs3/pubsys2/xml/en/procedures/28/28-008-002-tr.html)
Procedure Title	Drive Belt, Cooling Fan
Update Reason	Updated to include an inspect and re-use guideline for the drive belt tensioner.
Date	11-May-2017
Section	Section 9 - Drive Units
Procedure Number	28-009-032-tr (/qs3/pubsys2/xml/en/procedures/28/28-009-032-tr.html)
Procedure Title	Water Pump Drive Pulley
Update Reason	Added Inspect for Reuse and Pose steps.
Date	12-May-2017
Section	Section 9 - Drive Units
Procedure Number	28-009-004-tr (/qs3/pubsys2/xml/en/procedures/28/28-009-004-tr.html)
Procedure Title	Accessory Drive Pulley
Update Reason	This manual has been changed to make it more consistent with the Water Pump drive pulley 009-032. Repeated steps removed.
Date	12-May-2017
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	28-001-003-tr (/qs3/pubsys2/xml/en/procedures/28/28-001-003-tr.html)
Procedure Title	Accessory Drive Seal
Update Reason	This manual has been changed to ensure the terminology is consistent throughout qosol. Oil slinger has been amended to Keyway seal.

Date	08-Mar-2017
Section	Section 9 - Drive Units - Group 09
Procedure Number	28-009-011-tr (/qs3/pubsys2/xml/en/procedures/28/28-009-011-tr.html)
Procedure Title	Fuel Pump Drive
Update Reason	The fuel pump drive shaft measurements and tolerances were incorrect. This has been updated.

Date	30-May-2017
Section	Section 5 - Fuel System - Group 05
Procedure Number	28-005-036 (/qs3/pubsys2/xml/en/procedures/28/28-005-036.html)
Procedure Title	Fuel Pump Throttle Linkage
Update Reason	Procedure added to manual.

Date	17-Jul-2017
Section	Section 5 - Fuel System
Procedure Number	28-005-045-tr (/qs3/pubsys2/xml/en/procedures/28/28-005-045-tr.html)
Procedure Title	Fuel Lift Pump
Update Reason	Pressure sensor replaced with Compuccheck™ fitting.

Date	25-Aug-2017
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	28-001-016-tr (/qs3/pubsys2/xml/en/procedures/28/28-001-016-tr.html)
Procedure Title	Crankshaft
Update Reason	Updated Clean and Inspect for Reuse step.

Date	19-Sep-2017
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	28-001-043-tr (/qs3/pubsys2/xml/en/procedures/28/28-001-043-tr.html)
Procedure Title	Cylinder Liner
Update Reason	Corrected length of piston pin.
Date	19-Sep-2017
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	28-001-016-tr (/qs3/pubsys2/xml/en/procedures/28/28-001-016-tr.html)
Procedure Title	Crankshaft
Update Reason	The procedure was updated with clearer information when inspecting for reuse.
Date	29-Jan-2018
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	28-001-083-tr (/qs3/pubsys2/xml/en/procedures/28/28-001-083-tr.html)
Procedure Title	Idler Shaft
Update Reason	Updated with a second capscrew, additional information for identification and revised torque sequence.
Date	06-Feb-2018
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	28-001-010-tr (/qs3/pubsys2/xml/en/procedures/28/28-001-010-tr.html)
Procedure Title	Camshaft Bushings
Update Reason	Corrected the Camshaft Bushing Bore in the Block table.

Date	20-Feb-2018
Section	Section 6 - Injectors and Fuel Lines - Group 06
Procedure Number	28-006-003 (/qs3/pubsys2/xml/en/procedures/28/28-006-003.html)
Procedure Title	Air in Fuel
Update Reason	Added the test with Electronically Actuated Injector.

Date	15-Jun-2018
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021528-t027-tr (/qs3/pubsys2/xml/en/manual/4021528/4021528-t027-tr.html)
Procedure Title	Crankcase Gases (Blowby) Excessive
Update Reason	Corrected reference to Fault Codes 556, 1843, and 1844 in the TS tree.

Date	23-Jul-2018
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	28-001-027-tr (/qs3/pubsys2/xml/en/procedures/28/28-001-027-tr.html)
Procedure Title	Cylinder Block and Liner Seats
Update Reason	Revised the Upper Counter Bore diameter values.

Date	07-Nov-2018
Section	Section 7 - Lubricating Oil System - Group 07
Procedure Number	28-007-067-tr (/qs3/pubsys2/xml/en/procedures/28/28-007-067-tr.html)
Procedure Title	Eliminator™ Filter
Update Reason	Revised 'Disassemble', 'Clean and Inspect for Reuse', and 'Assemble' step to redirect to Bulletin 3666253.

Date	20-Nov-2018
Section	Section V - Maintenance Specifications
Procedure Number	28-018-024-tr (/qs3/pubsys2/xml/en/procedures/28/28-018-024-tr.html)
Procedure Title	Filter Specifications
Update Reason	Added Fuel and Water Seperator and Lube Centrifugal By-Pass Filter specifications.
Date	07-Jan-2019
Section	Section 13 - Electrical Equipment
Procedure Number	28-013-005-tr (/qs3/pubsys2/xml/en/procedures/28/28-013-005-tr.html)
Procedure Title	Charging System Alternator Drive Belt
Update Reason	Updated instruction for installing and removing the automatic belt tensioner.
Date	11-Feb-2019
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	28-001-043-tr (/qs3/pubsys2/xml/en/procedures/28/28-001-043-tr.html)
Procedure Title	Piston
Update Reason	Revised specifications for Piston Pin Length.
Date	11-Feb-2019
Section	Section 13 - Electrical Equipment
Procedure Number	28-013-003 (/qs3/pubsys2/xml/en/procedures/28/28-013-003.html)
Procedure Title	Alternator Bracket
Update Reason	Added automatic belt tensioner remove and install steps.

Date	15-Feb-2019
Section	Section 1 - Cylinder Block
Procedure Number	28-001-019-tr (/qs3/pubsys2/xml/en/procedures/28/28-001-019-tr.html)
Procedure Title	Crankshaft Gear, Front (Crankshaft Removed)
Update Reason	Revised outside diameter measurement for crankshaft.
Date	22-Feb-2019
Section	Section 1 - Cylinder Block
Procedure Number	28-001-026-tr (/qs3/pubsys2/xml/en/procedures/28/28-001-026-tr.html)
Procedure Title	Cylinder Block
Update Reason	Updated standard or oversize liners table.
Date	28-Mar-2019
Section	Section 9 - Drive Units - Group 09
Procedure Number	28-009-035-tr (/qs3/pubsys2/xml/en/procedures/28/28-009-035-tr.html)
Procedure Title	Engine Barring Device
Update Reason	Added to outline.
Date	03-May-2019
Section	Section 3 - Rocker Levers - Group 03
Procedure Number	28-003-013-tr (/qs3/pubsys2/xml/en/procedures/28/28-003-013-tr.html)
Procedure Title	Rocker Lever Housing
Update Reason	Edited to include additional steps in both the preliminary and finishing steps. These additional steps were added to remove pipework that can potentially get in the way when removing the housing in the field.

Date	13-May-2019
Section	Section 7 - Lubricating Oil System - Group 07
Procedure Number	28-007-045-tr (/qs3/pubsys2/xml/en/procedures/28/28-007-045-tr.html)
Procedure Title	Lubricating Oil Cooler Cover
Update Reason	Changed the torque sequence and torque value.
Date	02-Jul-2019
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	28-001-027-tr (/qs3/pubsys2/xml/en/procedures/28/28-001-027-tr.html)
Procedure Title	Cylinder Block and Liner Seats
Update Reason	Updated standard or oversize liners table.
Date	05-Aug-2019
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021528-t104-tr (/qs3/pubsys2/xml/en/manual/4021528/4021528-t104-tr.html)
Procedure Title	Lubricating Oil Pressure High
Update Reason	Added piston cooling nozzle node to tree.
Date	09-Sep-2019
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	28-001-083-tr (/qs3/pubsys2/xml/en/procedures/28/28-001-083-tr.html)
Procedure Title	Idler Shaft
Update Reason	Updated the torque angle from 60 degrees to 90 degrees.

Date	12-Sep-2019
Section	Section 10 - Air Intake System - Group 10
Procedure Number	28-010-008-tr (/qs3/pubsys2/xml/en/procedures/28/28-010-008-tr.html)
Procedure Title	Aftercooler Element
Update Reason	Removed reference to the gasket for the LTA aftercooler.
Date	16-Sep-2019
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	28-001-083-tr (/qs3/pubsys2/xml/en/procedures/28/28-001-083-tr.html)
Procedure Title	Idler Shaft
Update Reason	Updated graphics in Install step.
Date	23-Sep-2019
Section	Section 8 - Cooling System - Group 08
Procedure Number	28-008-002-tr (/qs3/pubsys2/xml/en/procedures/28/28-008-002-tr.html)
Procedure Title	Drive Belt, Cooling Fan
Update Reason	Updated to include more information for K50 K1800E when you operate the fan belt and then check its tenion.
Date	14-Oct-2019
Section	Section 1 - Cylinder Block
Procedure Number	28-001-083-tr (/qs3/pubsys2/xml/en/procedures/28/28-001-083-tr.html)
Procedure Title	Idler Shaft
Update Reason	Revised Note in Install to state second flat in a clockwise direction instead of first.

Date	08-Nov-2019
Section	Section F - Familiarization
Procedure Number	28-200-002-tr (/qs3/pubsys2/xml/en/procedures/28/28-200-002-tr.html)
Procedure Title	Flow Diagram, Lubricating Oil System
Update Reason	Reversed positions of oil flow arrows in graphic 07e00194 to indicate proper oil flow.
Date	18-Nov-2019
Section	Section 11 - Exhaust System - Group 11
Procedure Number	28-011-032-tr (/qs3/pubsys2/xml/en/procedures/28/28-011-032-tr.html)
Procedure Title	Heat Shield
Update Reason	Added procedure to manual.
Date	11-Dec-2019
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	28-001-083-tr (/qs3/pubsys2/xml/en/procedures/28/28-001-083-tr.html)
Procedure Title	Idler Shaft
Update Reason	Switched indicators for the water pump idler gear and right bank camshaft idler gear in the Remove step of the procedure.
Date	23-Dec-2019
Section	Section 6 - Injectors and Fuel Lines - Group 06
Procedure Number	28-006-013 (/qs3/pubsys2/xml/en/procedures/28/28-006-013.html)
Procedure Title	Fuel Drain Lines
Update Reason	Added stepblock to Remove step that describes how drain lines are removed.

Date	06-Jan-2020
Section	Section 6 - Injectors and Fuel Lines - Group 06
Procedure Number	28-006-017-tr (/qs3/pubsys2/xml/en/procedures/28/28-006-017-tr.html)
Procedure Title	Fuel Filter Head
Update Reason	Added stepblock to Assemble step that lists the torque value for QSK38 Engines with Gerotor Fuel Pump Drain Lines.

Date	06-Jan-2020
Section	Section 6 - Injectors and Fuel Lines - Group 06
Procedure Number	28-006-021-tr (/qs3/pubsys2/xml/en/procedures/28/28-006-021-tr.html)
Procedure Title	Fuel Manifold (Drain)
Update Reason	Added stepblock that describes the torque value for inserting an elbow fitting onto QSK38 and QSK50 Engines with Gerotor Fuel Pump Drain Lines.

Date	14-Jan-2020
Section	Section 6 - Injectors and Fuel Lines - Group 06
Procedure Number	28-006-024-tr (/qs3/pubsys2/xml/en/procedures/28/28-006-024-tr.html)
Procedure Title	Fuel Supply Lines
Update Reason	Added stepblocks to the Clean and Inspect for Reuse steps. Added stepblocks to Install steps.

Date	22-Jan-2020
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021528-t105-tr (/qs3/pubsys2/xml/en/manual/4021528/4021528-t105-tr.html)
Procedure Title	Lubricating Oil Pressure Low
Update Reason	Added procedure reference to 007-024.

Date	29-Jan-2020
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	28-001-058-tr (/qs3/pubsys2/xml/en/procedures/28/28-001-058-tr.html)
Procedure Title	Cylinder Block Counterbore
Update Reason	Revised the listed minimum and maximum specifications for the cylinder block reference diameter (upper counterbore) that applies to the 2.54 mm [0.100 in] Oversize Repair Sleeve.
Date	03-Feb-2020
Section	Section 14 - Engine Testing - Group 14
Procedure Number	28-022-001_14-tr (/qs3/pubsys2/xml/en/procedures/28/28-022-001_14-tr.html)
Procedure Title	Service Tools
Update Reason	Added cylinder diagnostic tool to Service Tools table.
Date	17-Mar-2020
Section	Section 6 - Injectors and Fuel Lines - Group 06
Procedure Number	28-006-017-tr (/qs3/pubsys2/xml/en/procedures/28/28-006-017-tr.html)
Procedure Title	Fuel Filter Head
Update Reason	Added procedure to manual.
Date	25-Mar-2020
Section	Section 14 - Engine Testing - Group 14
Procedure Number	00-014-012 (/qs3/pubsys2/xml/en/procedures/00/00-014-012.html)
Procedure Title	Engine Compression
Update Reason	Added additional information to the procedure.

Date	26-Mar-2020
Section	Section 12 - Compressed Air System - Group 12
Procedure Number	28-012-020 (/qs3/pubsys2/xml/en/procedures/28/28-012-020.html)
Procedure Title	Air Compressor (Oil Carryover)
Update Reason	Reconfigured procedure to commonize it.
Date	26-Mar-2020
Section	Section 11 - Exhaust System - Group 11
Procedure Number	28-011-007-tr (/qs3/pubsys2/xml/en/procedures/28/28-011-007-tr.html)
Procedure Title	Exhaust Manifold, Dry
Update Reason	Added stepblock to the Finishing Steps of the procedure.
Date	12-May-2020
Section	Section 5 - Fuel System - Group 05
Procedure Number	102-005-232 (/qs3/pubsys2/xml/en/procedures/102/102-005-232.html)
Procedure Title	Fuel Pump Pressurizing Assembly
Update Reason	Added Resisitance Check step to procedure.
Date	17-Jul-2020
Section	Section 0 - Product - Group 00
Procedure Number	102-000-006-tr-gc (/qs3/pubsys2/xml/en/procedures/102/102-000-006-tr-gc.html)
Procedure Title	Engine Storage - Short Term
Update Reason	Added procedure to manual.

Date	12-Aug-2020
Section	Section 5 - Fuel System - Group 05
Procedure Number	28-005-016-tr (/qs3/pubsys2/xml/en/procedures/28/28-005-016-tr.html)
Procedure Title	Fuel Pump
Update Reason	Added note to indicate the check cannot be carried out on later models of the pumps due to the weep hole being removed.
Date	09-Sep-2020
Section	Section 2 - Cylinder Head - Group 02
Procedure Number	28-002-004-tr (/qs3/pubsys2/xml/en/procedures/28/28-002-004-tr.html)
Procedure Title	Cylinder Head
Update Reason	Made minor corrections to procedure.
Date	16-Mar-2021
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	28-001-058-tr (/qs3/pubsys2/xml/en/procedures/28/28-001-058-tr.html)
Procedure Title	Cylinder Block Counterbore
Update Reason	Removed mention and graphics pertaining to service tool, Part Number 3375820, from the procedure.
Date	26-Mar-2021
Section	Section 16 - Mounting Adaptations - Group 16
Procedure Number	28-016-004-tr (/qs3/pubsys2/xml/en/procedures/28/28-016-004-tr.html)
Procedure Title	Flexplate
Update Reason	Added information regarding the insulated flexplate to the procedure.

Date	08-Apr-2021
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	28-001-024-tr (/qs3/pubsys2/xml/en/procedures/28/28-001-024-tr.html)
Procedure Title	Crankshaft Seal, Rear
Update Reason	Corrected listed part number for rear crankshaft seal replacer kit, from Part Number 4919752 to Part Number 5394933.

Date	08-Apr-2021
Section	Section 7 - Lubricating Oil System - Group 07
Procedure Number	28-007-063-tr (/qs3/pubsys2/xml/en/procedures/28/28-007-063-tr.html)
Procedure Title	Pre-Lubricating Oil Pump
Update Reason	Added additional necessary detail to General Information section. Added torque values for 3/8" capscrew.

Date	27-Apr-2021
Section	Section 3 - Rocker Levers - Group 03
Procedure Number	28-003-013-tr (/qs3/pubsys2/xml/en/procedures/28/28-003-013-tr.html)
Procedure Title	Rocker Lever Housing
Update Reason	Removed any mention of Loctite™ 5910 sealant in procedure as it is no longer used.

Date	25-May-2021
Section	Section 7 - Lubricating Oil System - Group 07
Procedure Number	28-007-013-tr (/qs3/pubsys2/xml/en/procedures/28/28-007-013-tr.html)
Procedure Title	Lubricating Oil Filter (Spin-On)
Update Reason	Corrected listed measurements for Maximum Oil Filter Pressure Differential to reflect accurate values.

Date	02-Jun-2021
Section	Section 7 - Lubricating Oil System - Group 07
Procedure Number	28-007-061-tr (/qs3/pubsys2/xml/en/procedures/28/28-007-061-tr.html)
Procedure Title	Oil Transfer Connection
Update Reason	Revised wording to ensure that both faces of the oil transfer connection are being cleaned and inspected for reuse.
Date	10-Jun-2021
Section	Section 7 - Lubricating Oil System - Group 07
Procedure Number	28-007-040-tr (/qs3/pubsys2/xml/en/procedures/28/28-007-040-tr.html)
Procedure Title	Lubricating Oil Transfer Tube
Update Reason	Added instruction to install the outward embossed face towards the transfer block face and the flat back face of the gasket towards the engine block.
Date	10-Jun-2021
Section	Section 6 - Injectors and Fuel Lines - Group 06
Procedure Number	28-006-025-tr (/qs3/pubsys2/xml/en/procedures/28/28-006-025-tr.html)
Procedure Title	Static Injection Timing
Update Reason	Corrected values listed in Table C.
Date	17-Jun-2021
Section	Section 8 - Cooling System - Group 08
Procedure Number	28-008-002-tr (/qs3/pubsys2/xml/en/procedures/28/28-008-002-tr.html)
Procedure Title	Drive Belt, Cooling Fan
Update Reason	Corrected graphic to show a breaker bar instead of an adjustable wrench.

Date	29-Jun-2021
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	28-001-058-tr (/qs3/pubsys2/xml/en/procedures/28/28-001-058-tr.html)
Procedure Title	Cylinder Block Counterbore
Update Reason	Corrected listed Interference Fit on Diameter for both the Straight Sleeve and "L" Shaped Sleeve.
Date	09-Jul-2021
Section	Section 6 - Injectors and Fuel Lines - Group 06
Procedure Number	28-006-012 (/qs3/pubsys2/xml/en/procedures/28/28-006-012.html)
Procedure Title	Fuel Drain Line Restriction
Update Reason	Added steps for the fuel drain line with an electronically actuated injector.
Date	21-Jul-2021
Section	Section 19 - Electronic Controls - Group 19
Procedure Number	122-019-450 (/qs3/pubsys2/xml/en/procedures/122/122-019-450.html)
Procedure Title	Exhaust Gas Temperature Sensor Converter
Update Reason	Added procedure to manual.
Date	11-Aug-2021
Section	Group 10 - Air Intake System - Group 10
Procedure Number	28-010-033-tr (/qs3/pubsys2/xml/en/procedures/28/28-010-033-tr.html)
Procedure Title	Turbocharger
Update Reason	Updated minimum and maximum specifications for Compressor Impeller Axial End Clearance and Compressor Impeller Radial Clearance to include Holset® HX82/HX83/Holset® HE800.

Date	25-Aug-2021
Section	Section 16 - Mounting Adaptations - Group 16
Procedure Number	28-016-006-tr (/qs3/pubsys2/xml/en/procedures/28/28-016-006-tr.html)
Procedure Title	Flywheel Housing
Update Reason	Revised maximum runout specifications shown in the table(s) Flywheel Housing - Radial Run Out and Flywheel Housing - Face Run Out to reflect correct values.
Date	25-Aug-2021
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	28-001-016-tr (/qs3/pubsys2/xml/en/procedures/28/28-001-016-tr.html)
Procedure Title	Crankshaft
Update Reason	Updated graphic to show where the crankshaft should be supported on the V-blocks.
Date	11-Oct-2021
Section	Section 8 - Cooling System - Group 08
Procedure Number	28-008-002-tr (/qs3/pubsys2/xml/en/procedures/28/28-008-002-tr.html)
Procedure Title	Drive Belt, Cooling Fan
Update Reason	Added step to show installation of the belt tension gauge.
Date	11-Oct-2021
Section	Section 6 - Injectors and Fuel Lines - Group 06
Procedure Number	102-006-061 (/qs3/pubsys2/xml/en/procedures/102/102-006-061.html)
Procedure Title	Fuel Pressure Relief Valve
Update Reason	Added mention of graduated cylinder into the procedure.

Date	11-Oct-2021
Section	Section 7 - Lubricating Oil System - Group 07
Procedure Number	28-007-017 (/qs3/pubsys2/xml/en/procedures/28/28-007-017.html)
Procedure Title	Lubricating Oil Filter Head (Remote-Mounted)
Update Reason	Revised Disassemble and Assemble steps.
Date	16-Aug-2022
Section	Section V - Maintenance Specifications
Procedure Number	28-018-024-tr (/qs3/pubsys2/xml/en/procedures/28/28-018-024-tr.html)
Procedure Title	Filter Specifications
Update Reason	Added additional Cummins® and Flleetguard® part number for Fuel Filter, Stage 2 (NanoNet™ media) with Electronically Actuated Injectors.
Date	20-Sep-2022
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	28-022-001_01-tr (/qs3/pubsys2/xml/en/procedures/28/28-022-001_01-tr.html)
Procedure Title	Service Tools
Update Reason	Added seal installer, Part Number 5573467 and accompanying graphic to tools list.
Date	26-Sep-2022
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	28-001-031-tr (/qs3/pubsys2/xml/en/procedures/28/28-001-031-tr.html)
Procedure Title	Gear Cover, Front
Update Reason	Removed mention of using a mandrel to install the accessory drive seal to use the seal installer, Part Number 5573467 instead.

Date	28-Oct-2022
Section	Section 6 - Injectors and Fuel Lines - Group 06
Procedure Number	28-006-037-tr (/qs3/pubsys2/xml/en/procedures/28/28-006-037-tr.html)
Procedure Title	STC Oil Control Valve (Mechanical)
Update Reason	Added information to table for Valve Part Number 3633381.

Date	28-Oct-2022
Section	Section 0 - Product - Group 00
Procedure Number	28-000-001-tr (/qs3/pubsys2/xml/en/procedures/28/28-000-001-tr.html)
Procedure Title	Engine Removal
Update Reason	Corrected part number listed for engine lifting fixture, from Part Number 3375109 to 3162364.

Date	28-Oct-2022
Section	Section 0 - Product - Group 00
Procedure Number	28-000-002-tr (/qs3/pubsys2/xml/en/procedures/28/28-000-002-tr.html)
Procedure Title	Engine Installation
Update Reason	Corrected part number listed for engine lifting fixture, from Part Number 3375109 to 3162364.

Date	02-Jan-2023
Section	Section 6 - Injectors and Fuel Lines - Group 06
Procedure Number	28-006-003 (/qs3/pubsys2/xml/en/procedures/28/28-006-003.html)
Procedure Title	Air in Fuel
Update Reason	Deleted and added text to revise actions to take if an air leak is found.

Date	04-Jan-2023
Section	Section 7 - Lubricating Oil System - Group 07
Procedure Number	28-007-125 (/qs3/pubsys2/xml/en/procedures/28/28-007-125.html)
Procedure Title	Fan Clutch Lubricating Oil Manifold
Update Reason	Added procedure to manual.
Date	27-Jan-2023
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	28-001-054-tr (/qs3/pubsys2/xml/en/procedures/28/28-001-054-tr.html)
Procedure Title	Piston and Connecting Rod Assembly
Update Reason	Added text to indicate there may be an arrow or the letter F on the bottom of the piston pin bore to indicate the front of the engine.
Date	21-Feb-2023
Section	Section 2 - Cylinder Head - Group 02
Procedure Number	28-002-004-tr (/qs3/pubsys2/xml/en/procedures/28/28-002-004-tr.html)
Procedure Title	Cylinder Head
Update Reason	Updated Assemble step to display proper parts list.
Date	26-Apr-2023
Section	Section 7 - Lubricating Oil System - Group 07
Procedure Number	28-007-125 (/qs3/pubsys2/xml/en/procedures/28/28-007-125.html)
Procedure Title	Fan Clutch Lubricating Oil Manifold
Update Reason	Corrected final Install stepblock to include torque sequence.

Date	26-Apr-2023
Section	Section 16 - Mounting Adaptations - Group 16
Procedure Number	28-016-004-tr (/qs3/pubsys2/xml/en/procedures/28/28-016-004-tr.html)
Procedure Title	Flexplate
Update Reason	Added to Measure step that Cummins recommends to install a flywheel whenever inspecting the flywheel face and outer runouts. In situations that do not allow for installation of a flywheel and the only method of gaining flywheel face and outer runouts is via the flexplate option.
Date	15-May-2023
Section	Section 7 - Lubricating Oil System - Group 07
Procedure Number	28-007-015-tr (/qs3/pubsys2/xml/en/procedures/28/28-007-015-tr.html)
Procedure Title	Lubricating Oil Filter Head
Update Reason	Added information for duplex type filter head to procedure.
Date	24-May-2023
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	28-001-052-tr (/qs3/pubsys2/xml/en/procedures/28/28-001-052-tr.html)
Procedure Title	Vibration Damper, Viscous
Update Reason	Updated specifications for Maximum Vibration Damper Thickness listed in Clean and Inspect for Reuse step.
Date	13-Sep-2023
Section	Section 3 - Rocker Levers - Group 03
Procedure Number	28-003-006-tr (/qs3/pubsys2/xml/en/procedures/28/28-003-006-tr.html)
Procedure Title	Overhead Set (OBC)
Update Reason	Table misaligned in procedure, edited procedure to correct table.

Date	23-Oct-2023
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	28-001-006-tr (/qs3/pubsys2/xml/en/procedures/28/28-001-006-tr.html)
Procedure Title	Bearings, Main
Update Reason	Torque value corrected in second to last Install step of procedure.
Date	01-Apr-2025
Section	Section 6 - Injectors and Fuel Lines - Group 06
Procedure Number	28-006-037-tr (/qs3/pubsys2/xml/en/procedures/28/28-006-037-tr.html)
Procedure Title	STC Oil Control Valve (Mechanical)
Update Reason	Updated values to match accompanying service bulletins so that specifications and part numbers aligned.
Date	01-Oct-2025
Section	Section 6 - Injectors and Fuel Lines - Group 06
Procedure Number	28-006-037-tr (/qs3/pubsys2/xml/en/procedures/28/28-006-037-tr.html)
Procedure Title	STC Oil Control Valve (Mechanical)
Update Reason	Reversed valve part numbers 3060609 and 3063022 positions within the table to verify rest of the information and specifications match appropriately.
Date	02-Mar-2026
Section	Section 5 - Fuel System - Group 05
Procedure Number	28-005-236 (/qs3/pubsys2/xml/en/procedures/28/28-005-236.html)
Procedure Title	Fuel System Diagnostics
Update Reason	Added procedure to manual.

Date	13-May-2026
Section	Section 11 - Exhaust System - Group 11
Procedure Number	28-011-007-tr (/qs3/pubsys2/xml/en/procedures/28/28-011-007-tr.html)
Procedure Title	Exhaust Manifold, Dry
Update Reason	Updated procedure due to lock plate part number changes 3043912 to 3628969.

Last Modified: 10-Jun-2026
